

PLaser

Physics-Informed Neural Network Diode Laser EDA Application

A Real-Time Parametric Optimization & Longitudinal Profiling Platform

USER MANUAL & REFERENCE GUIDE

TARGET AUDIENCE:

Laser Cavity Designers, Optoelectronic Engineers & Research Collaborators

AUTHOR:

Zhenwen Wan (AI + Semiconductor Simulation Expert)

1. SCOPE, METHOD & INSTALLATION GUIDE

1.1 Service Scope & Methodology

PLaser provides a next-generation Electronic Design Automation (EDA) interface for edge-emitting telecom diode lasers. By training a Physics-Informed Neural Network (PINN) on a multi-physics hierarchical grid, PLaser replaces slow iterative differential equation solvers with a 5 ms real-time surrogate. The methodology couples a 2D transverse Finite Element Method (Elmer FEM) for electro-thermal optics with a 1D longitudinal cavity rate equations shooting solver. The neural net enforces local carrier rate balance as a physical constraint during backpropagation, securing structural and physical consistency even on unseen test configurations.

1.2 System Requirements & Path Constraints

- Operating System: Windows 10/11 or Linux.
- Python Environment: Python 3.9 - 3.12 (64-bit).
- Path Limit Alert: Install the PLaser application in a directory with a short path length (e.g., C:\PLaser or C:\Users\Public\PLaser). Long directories can trigger Windows path limitations (WinError 206) when installing PyTorch libraries.

1.3 Step-by-Step Installation

To configure your local environment, launch PowerShell or your preferred terminal and run the following command sequence:

1. Clone the project files: `git clone https://github.com/ZhenwenWan/PLaser.git`
2. Create and activate a clean virtual environment: `python -m venv .venv .venv\Scripts\Activate.ps1`
3. Install dependencies (CPU-optimized PyTorch & Streamlit dashboard): `pip install -r requirements.txt`

2. RUNNING, VISUALIZING & EXPORTING RESULTS

2.1 Launching the Application Suite

PLaser provides three execution commands corresponding to your workflow stage: •
Generate New Dataset: If you want to expand sweeps beyond the preloaded 1,500 samples: `python generate_dataset.py` • Train surrogate from scratch: To retrain PyTorch PINN model weights: `python train_pinn.py` • Run Interactive Dashboard: To launch the user application interface: `python -m streamlit run app.py`

2.2 Real-time Visualization & Spatial Hole Burning

The web-based dashboard is split into two logical panels for ease of use: • Left Sidebar (Design Controls): Drag sliders to tweak Left Reflectivity (R1), Right Reflectivity (R2), Cavity Length (L), Operating Temperature (T0), and Injection Current. Changes take effect in 5 ms. • Right Main Panel (Lasing Performance): Displays real-time scalar metrics (Output Power, efficiency, current) and physical profiles. Setting asymmetric facet coatings (e.g., $R1=0.9$, $R2=0.05$) clearly depicts Spatial Hole Burning (SHB)—visible as an asymmetric carrier density dip $N(z)$ near the emitting facet.

2.3 Exporting Results

Users can export design results in three ways: 1. Chart Images: Hover over the upper-right corner of either the Optical Power or Carrier Density charts and click the camera icon to save the plot as a PNG/SVG file. 2. Page Reports: Use your browser's Print function (Ctrl+P) to export the dashboard layout to a PDF. 3. Programmatic Batch Export: Import the PINNSurrogate class in Python to sweep parameters and dump predictions into CSV or Excel files automatically.

3. PHYSICAL VALIDATION & PERFORMANCE METRICS

3.1 Predictive Accuracy vs. High-Fidelity Solvers

PLaser's PINN surrogate has been validated against unseen parameter sets. The R^2 scores demonstrate near-perfect alignment: Optical Output Power (0.998), Wall-Plug Efficiency (0.997), and Total Terminal Current (0.999). This validation proves that the network successfully learns the highly nonlinear electro-optical thresholds without loss of generalization.

3.2 Execution Speed & Latency Comparison

Simulator Method	Execution Latency	Accuracy (R^2)	Optimization Role
Conventional 2D Elmer FEM	12 - 25 seconds	1.000 (Reference)	Transverse Mode Baseline
Quasi-3D (2.5D) shooting solver	1.2 - 2.8 seconds	1.000 (Reference)	Longitudinal Sweeps
PLaser PINN Surrogate	< 5 milliseconds	> 0.998	Real-Time Sweeps / EDA

3.3 Custom Physics-Informed Training Loss Convergence

During training, the custom physics loss function enforces the continuity rate equations by evaluating gradients along multiple longitudinal slices. The loss history chart below demonstrates a clean 6-order-of-magnitude convergence over 150 epochs, proving physical compliance.

